Mathematics - Probability and Statistics

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Syllabus

- **Session 1:** Why Study Statistics?
- **Session 2:** Graphical Representation
- **Session 3:** Descriptive Statistics
- **Session 4:** Boxplots & Normal Distribution
- **Session 5:** Basic Probability & Counting Rules
- **Session 6:** Conditional Probability & Independence
- **Session 7:** Total Probability & Bayes' Theorem
- **Session 8:** Discrete Random Variables
- **Session 9:** Continuous Random Variables
- **Session 10:** Joint Distributions & Correlation
- **Session 11: Exercices/Summary**
- **Session 12:** Central Limit Theorem & Simulation
- **Session 13:** Point Estimation & MLE
- **Session 14:** Confidence Intervals
- **Session 15:** Hypothesis Testing – Basics

Continuous Random Variables

Session 9

Continuous Random Variables

Session 9

Objectives:

- Determine probabilities from probability density functions
- Determine probabilities from cumulative distribution functions and cumulative distribution functions from probability density functions, and the reverse
- Calculate means and variances for continuous random variables
- Understand the assumptions for some common continuous probability distributions

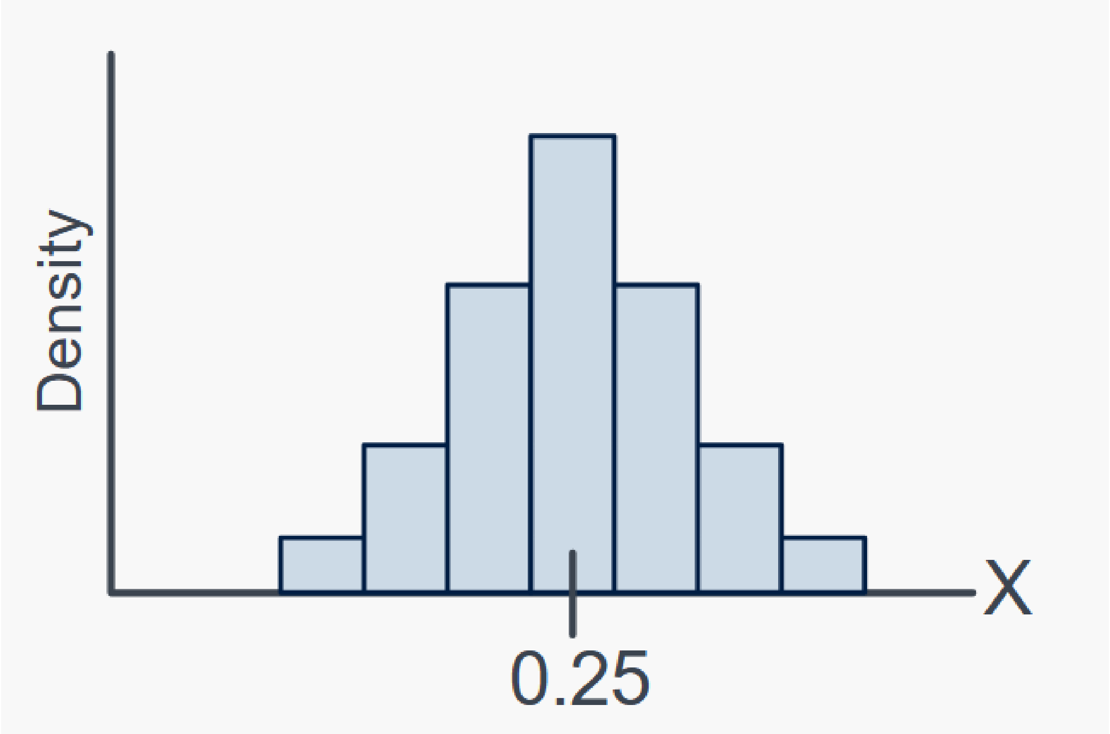
Continuous Random Variables

A **continuous random variable** differs from a discrete random variable in that it takes on an uncountably infinite number of possible outcomes.

For example: the height (in meters) of a randomly selected tree

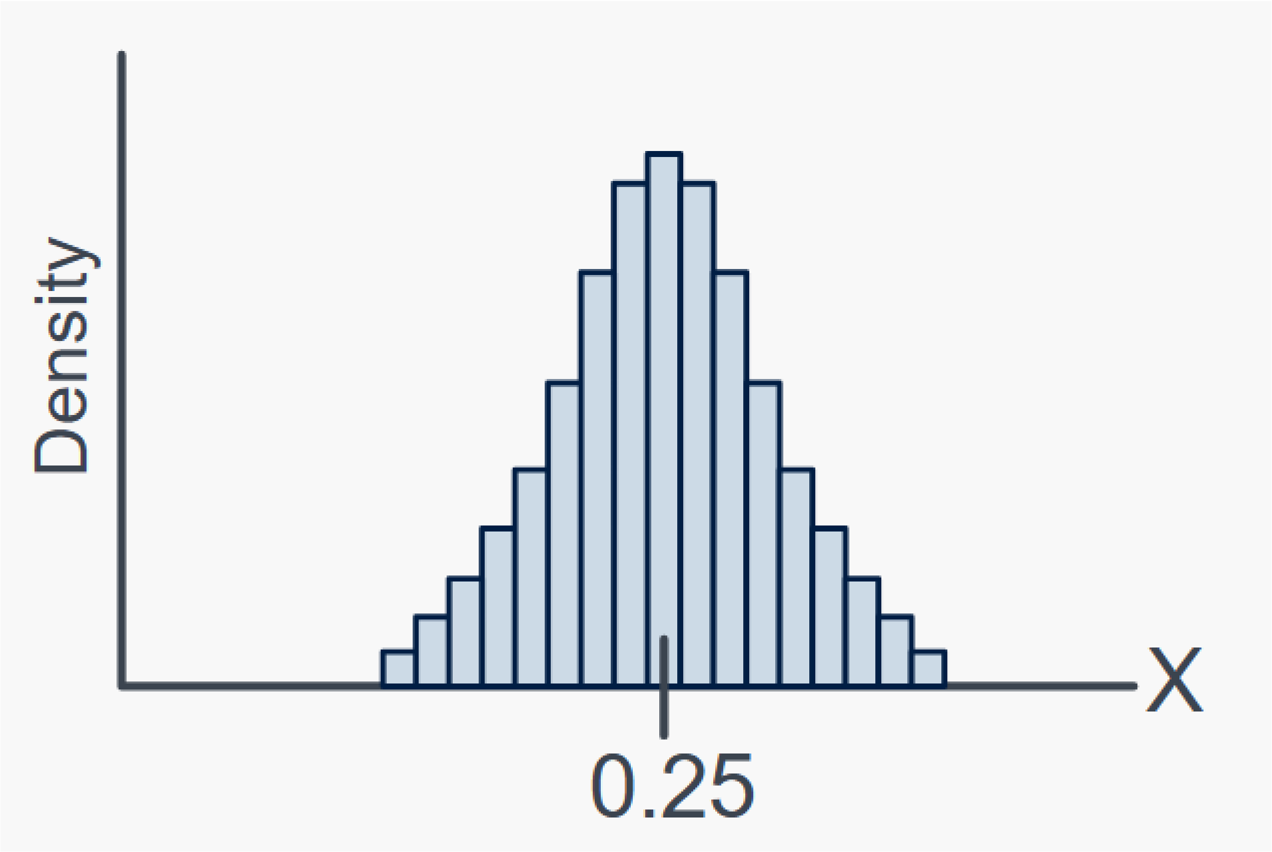
Discrete to continous

Imagine, the weight of a banana. You find this distribution between 0.18 and 0.32



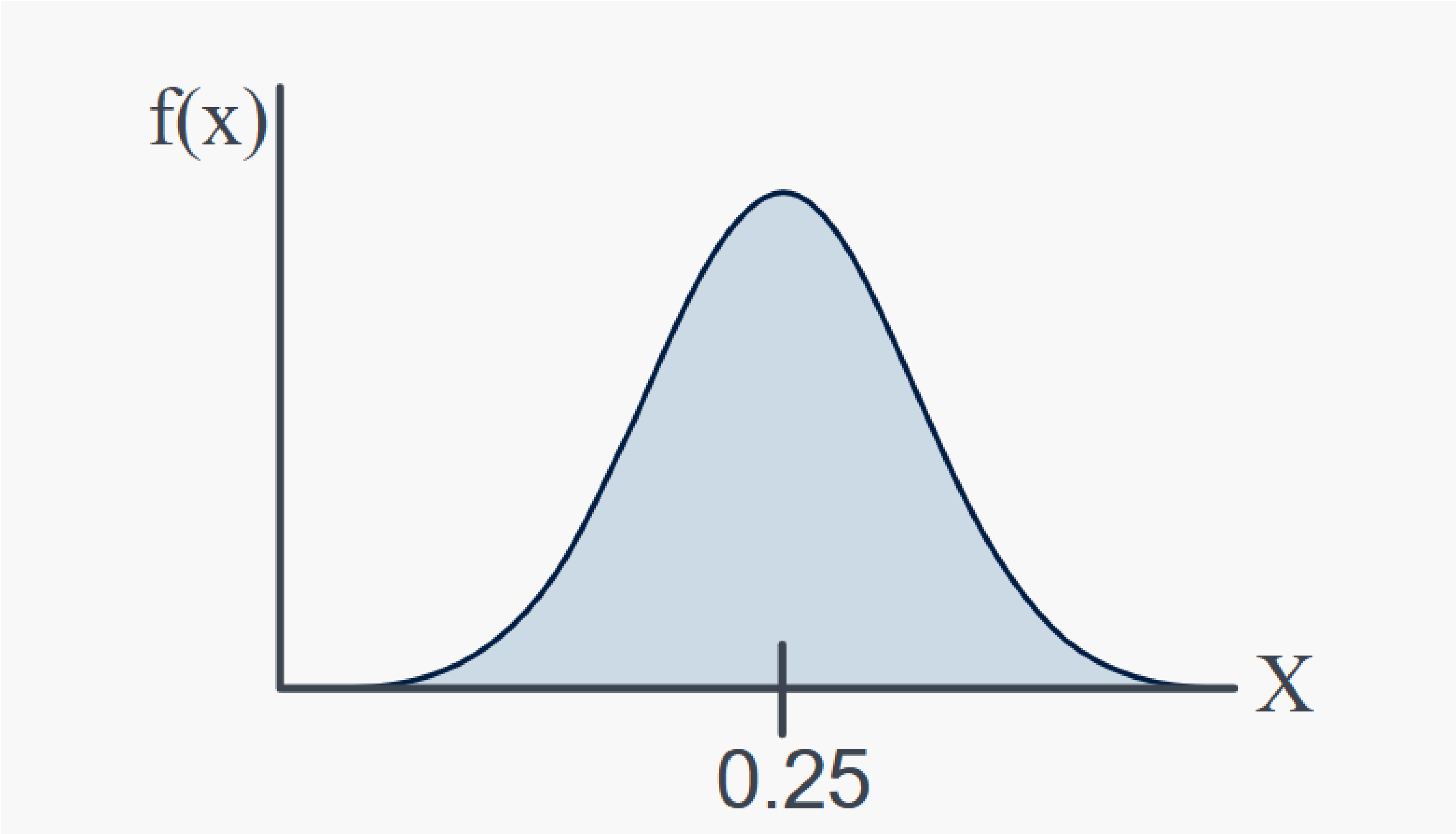
Discrete to continous

Now, you can decrease the bin size.



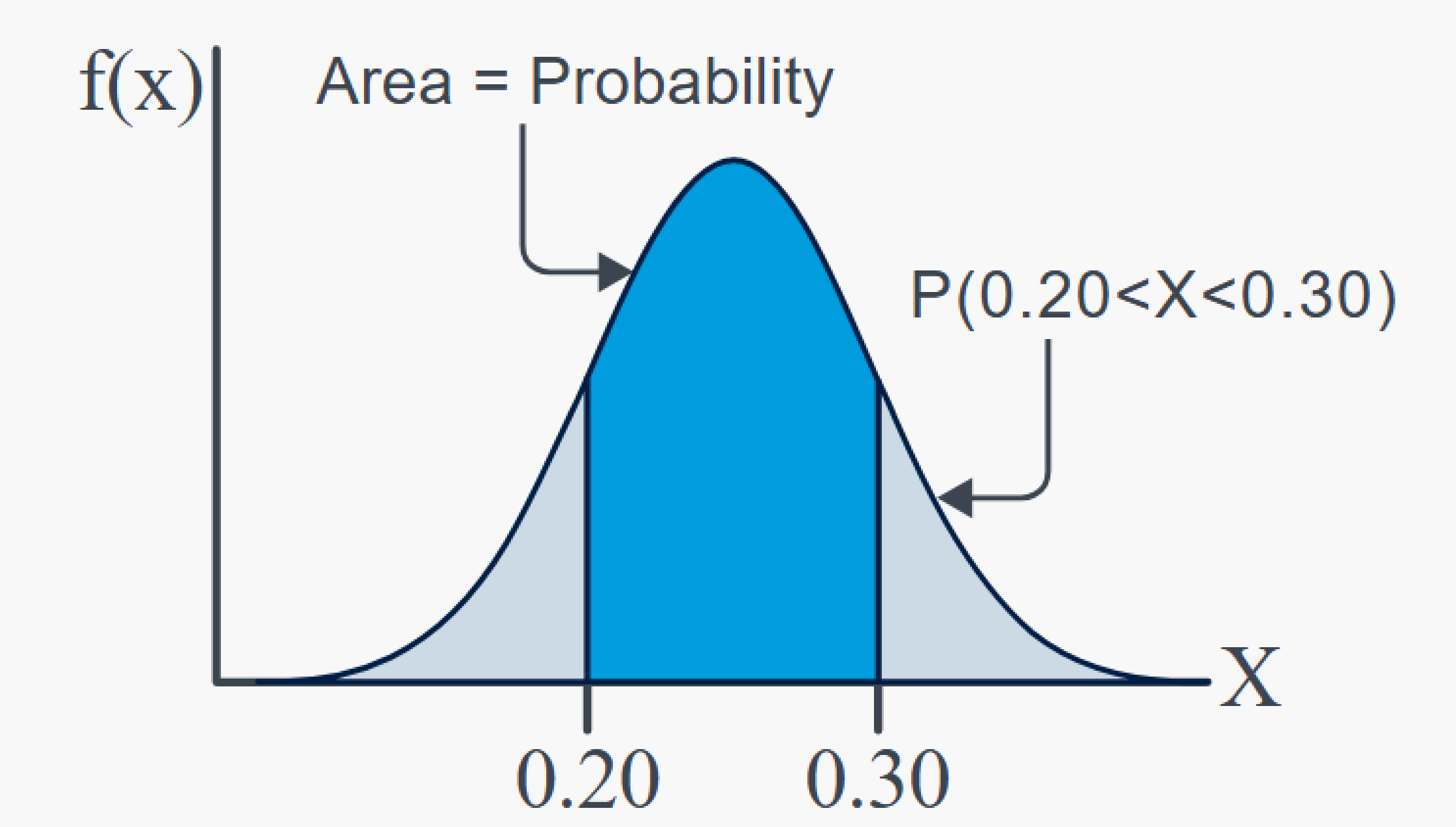
Discrete to continous

If you decrease again, we obtain a function f(x): **probability density function**



Discrete to continous

So if you want the probabilité that the weight is between 0.2 and 0.3



So what happens if you want P(X=x).

You can not have the area in 1 point!!!

P(X=x)=0 for x∈R

Discrete to continous

Definition:

For a continuous random variable X, a **probability density function** is a function such that

(1) $f(x) \geq 0$

(2) $\displaystyle\int_{-\infty}^{\infty} f(x)\,dx = 1$

(3) $P(a \leq X \leq b) = \displaystyle\int_{a}^{b} f(x)\,dx =$ area under $f(x)$ from a to b

for any a and b

PDF

Example: Let X be a continuous random variable whose probability density function is:

$f(x)=3x^2$ for $0<x<1$

First note: P(X=x) is different f(x). If you take x=0.9 you have f(0.9)=2.43 which is clearly not a probability!!!

The the probability is the area!!

$P(X=x)=\int_0^1 3x^2 = [x^3]_0^1=1$

Properties: Similarly, because each point has zero probability, one need not distinguish between inequalities such as $<$ or $\leq$ for continuous random variables.

If X is a **continuous random variable,** for any x_1 and x_2,

$$P(x_1 \leq X \leq x_2) = P(x_1 < X \leq x_2) = P(x_1 \leq X < x_2) = P(x_1 < X < x_2)$$

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Exercice: Let the continuous random variable X denote the current measured in a thin copper wire in milliamperes. Assume that the range of X is [0, 20 mA], and assume that the probability density function of X is $f(x)=0.05$ for X in [0, 20 mA]

What is the probability that a current measurement is less than 10 milliamperes?

PDF

Solution: Let the continuous random variable X denote the current measured in a thin copper wire in milliamperes. Assume that the range of X is [0, 20 mA], and assume that the probability density function of X is $f(x)=0.05$ for X in [0, 20 mA]

What is the probability that a current measurement is less than 10 milliamperes?

PDF

Exercice:
The probability density function of the net weight in pounds of a packaged chemical herbicide is $f(x)=2$ for $49.75<x<50.25$ pounds.

(a) Determine the probability that a package weighs more than 50 pounds.

(b) How much chemical is contained in 90% of all packages?

PDF

Solution:
The probability density function of the net weight in pounds of a packaged chemical herbicide is $f(x)=2$ for $49.75<x<50.25$ pounds.

(a) Determine the probability that a package weighs more than 50 pounds.

$P(X>50) = \int_{50}^{50.25} 2dx = 2[50.25-50]=0.5=50\%$

(b) How much chemical is contained in 90% of all packages?

$P(X<x1)=2[x1-49.75]=0.9$ therefore, $x1=50.2$

CDF

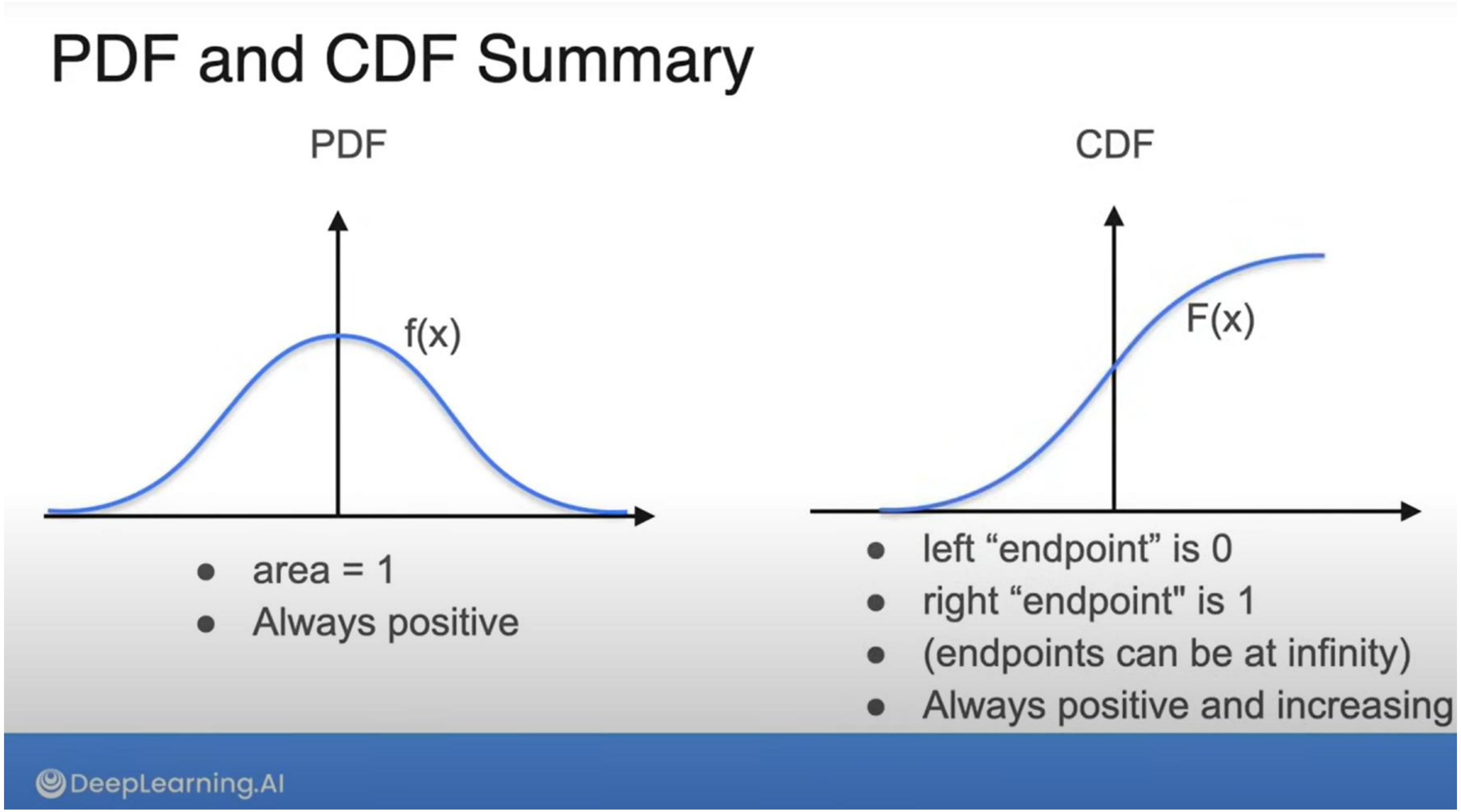
Definition:

The **cumulative distribution function** of a continuous random variable X is

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(u)\, du$$

for $-\infty < x < \infty$.

Summary PDF/CDF



CDF

Example: The time until a chemical reaction is complete (in milliseconds) is approximated by the cumulative distribution function

$$F(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-0.01x} & 0 \leq x \end{cases}$$

- Determine the probability density function of X.
- What proportion of reactions is complete within 200 milliseconds

CDF

Solution: The time until a chemical reaction is complete (in milliseconds) is approximated by the cumulative distribution function

$$F(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-0.01x} & 0 \leq x \end{cases}$$

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(u)\, du$$

- Determine the probability density function of X.

The probability density function is the derivative of F(x)

$$f(x) = \begin{cases} 0 & x < 0 \\ 0.01e^{-0.01x} & 0 \leq x \end{cases}$$

- What proportion of reactions is complete within 200 milliseconds

$P(X<200)=F(200)=1-e^{-2}=0.86$

CDF

Example: The probability density function of the time customers arrive at a terminal (in minutes after 8:00 A.M.) is $f(x)= e^{(-x/10)}/10$ for $0 < x$. Determine the probability that

(a) The first customer arrives by 9:00 A.M.
(b) The first customer arrives between 8:15 A.M. and 8:30 A.M.

CDF

Solution: The probability density function of the time customers arrive at a terminal (in minutes after 8:00 A.M.) is $f(x)= e^{(-x/10)}/10$ for $0 < x$. Determine the probability that

(a) The first customer arrives by 9:00 A.M.

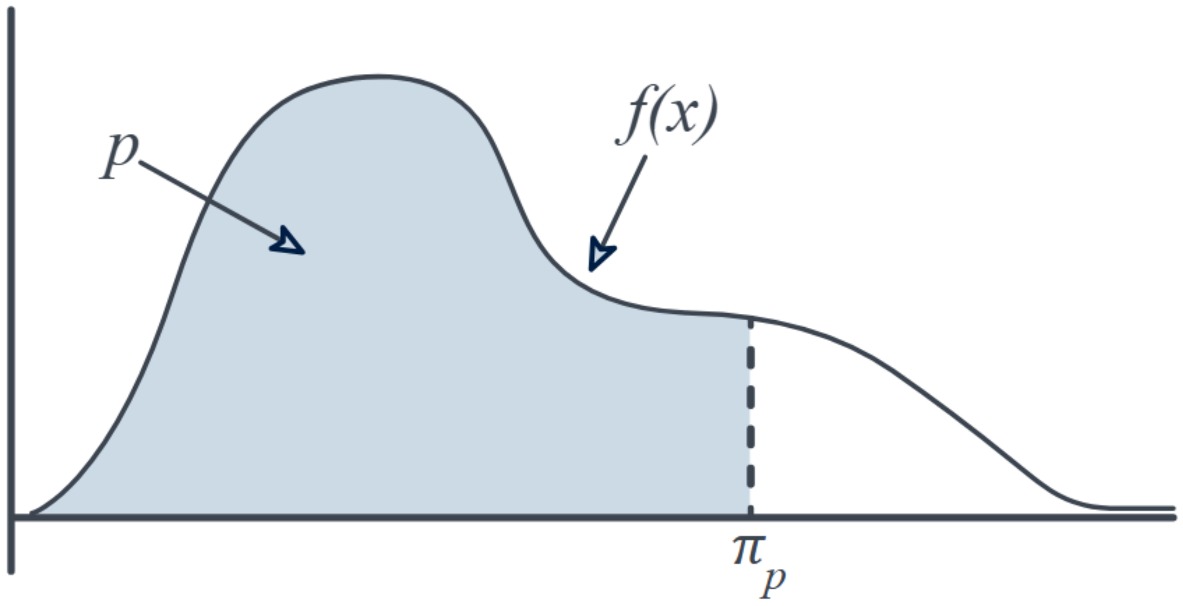
x=60 (1h after 8am=60min)
$P(X<60)= 1/10\int_0^{60} e^{(-x/10)}\,dx$ =-[$e^{(-60/10)}$- $e^{(-0/10)}$]=[1-e^{-6}]= 0.997

(b) The first customer arrives between 8:15 A.M. and 8:30 A.M.

x1=15,x2=30
$P(15<X<30)= 1/10\int_{15}^{30} e^{(-x/10)}\,dx$ =-[$e^{(-30/10)}$- $e^{(-15/10)}$]=0.173

CDF link with percentiles

Definition: If X is a continuous random variable, then the $(100p)^{th}$ **percentile** is a number π_p such that the area under f(x) and to the left of π_p is p.



CDF link with percentiles

Example: Let be X a continuous random variable with the following probability density function:
$f(x)=1/2$ for $0<x<2$.

What is the first quartile, median, and third quartile of X?

CDF link with percentiles

Solution: Let be X a continuous random variable with the following probability density function:
f(x)=1/2 for 0<x<2.

What is the first quartile, median, and third quartile of X?

F(x)= $1/2\int_0^x \quad dx = 0.5 *$[x- 0]=1/2x

Therefore 0.25=F($\pi_{0.25}$)=1/2* $\pi_{0.25}$

$\pi_{0.25}$ =0.5

Same 0.5= F($\pi_{0.50}$)=1/2* $\pi_{0.5}$ therefore $\pi_{0.50}$ =0.5*2=1

Same 0.75= F($\pi_{0.}$ 75)=1/2* $\pi_{0.75}$ therefore $\pi_{0.75}$ =0.75*2=1.5

CDF link with percentiles

Example: Let be X a continuous random variable with the following probability density function:
$f(x)=1/2(x+1)$ for $-1<x<1$.

What is the 64^{th} percentile of X?

CDF link with percentiles

Solution: Let be X a continuous random variable with the following probability density function:
$f(x)=1/2(x+1)$ for $-1<x<1$.

What is the 64^{th} percentile of X?

$F(x)= 1/2\int_{-1}^{x} x+1\,dx = 0.5 *[0.5x^2+x-(0.5-1)]=1/4(x^2+2x+1)=0.25(x+1)^2$

$0.64= F(\pi_{0.64})=0.25* (\pi_{0.64}+1)^2$ therefore $\pi_{0.64}=\text{sqrt}(0.64*4)-1=0.6$ because it should be between -1 and 1

Mean and standard deviation

The mean and variance can also be defined for a continuous random variable. Integration replaces summation in the discrete definitions.

Definition:

Suppose X is a continuous random variable with probability density function $f(x)$. The **mean** or **expected value** of X, denoted as μ or $E(X)$, is

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x)\, dx$$

The **variance** of X, denoted as $V(X)$ or σ^2, is

$$\sigma^2 = V(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)\, dx = \int_{-\infty}^{\infty} x^2 f(x)\, dx - \mu^2$$

The **standard deviation** of X is $\sigma = \sqrt{\sigma^2}$.

Mean and standard deviation

Examples: Let the continuous random variable X denote the current measured in a thin copper wire in milliamperes. Assume that the range of X is [0, 20 mA], and assume that the probability density function of X is $f(x)=0.05$ for $0<x<20$.

Derive the mean and the variance.

Mean and standard deviation

Solution: Let the continuous random variable X denote the current measured in a thin copper wire in milliamperes. Assume that the range of X is [0, 20 mA], and assume that the probability density function of X is f(x)=0.05 for 0<x<20.

Derive the mean and the variance.

E[X]=$\int_0^{20} xf(x)dx = \int_0^{20} 0.05x\, dx = 0.5 * 0.05 * 20^2$ =10

Var[X]=$\int_0^{20} x^2 f(x)dx$ $-$E^2[X]= $\int_0^{20} 0.05x^2\, dx - 10^2 = \frac{1}{3} * 0.05 * (20)^3$ -10^2=33.3

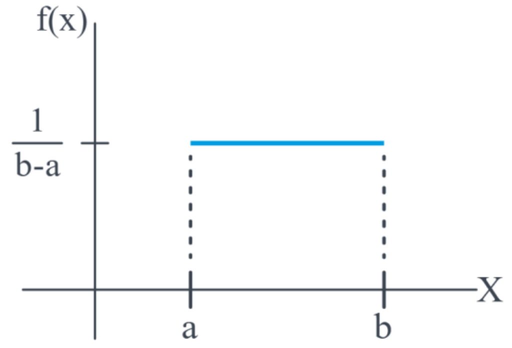
Important distribution: Uniform

Definition:

A continuous random variable X with probability density function

$$f(x) = 1/(b - a), \qquad a \leq x \leq b$$

is a **continuous uniform random variable.**

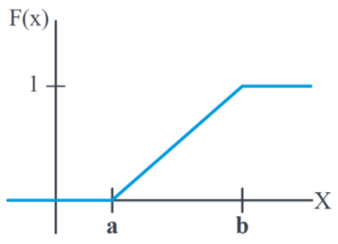


If X is a continuous uniform random variable over $a \leq x \leq b$,

$$\mu = E(X) = \frac{(a + b)}{2} \quad \text{and} \quad \sigma^2 = V(X) = \frac{(b - a)^2}{12}$$

Therefore, the complete description of the cumulative distribution function of a continuous uniform random variable is

$$F(x) = \begin{cases} 0 & x < a \\ (x - a)/(b - a) & a \leq x < b \\ 1 & b \leq x \end{cases}$$



Important distribution: Uniform

Exercice: Suppose X has a continuous uniform distribution over the interval [1.5, 5.5].

(a) Determine the mean, variance, and standard deviation of X.

(b) What is $P(X<2.5)$?

(c) Determine the cumulative distribution function.

Important distribution: Uniform

Solution: Suppose X has a continuous uniform distribution over the interval [1.5, 5.5].

(a) Determine the mean, variance, and standard deviation of X.

E[X]=(a+b)/2 =3.5
Var[X]=$(b\text{-}a)^2$/12=4/3 therefore std[X]=sqrt(4/3)=2/sqrt(3)

(b) What is P(X<2.5) ?

P(X<2.5)= $\int_{1.5}^{2.5} f(x)\ with\ f(x)$ =1/(b-a) therefore: P(X<2.5)=(2.5-a)/(b-a)=1/4

(c) Determine the cumulative distribution function.

$$F(x) = \begin{cases} 0, & x < 1.5, \\ \dfrac{x - 1.5}{4}, & 1.5 \leq x \leq 5.5, \\ 1, & x > 5.5. \end{cases}$$

Important distribution: Normal

It is the most widely used model for the distribution of a random.

Definition:

A random variable X with probability density function

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\mu)^2}{2\sigma^2}} \qquad -\infty < x < \infty$$

is a **normal random variable** with parameters μ, where $-\infty < \mu < \infty$, and $\sigma > 0$. Also,

$$E(X) = \mu \quad \text{and} \quad V(X) = \sigma^2$$

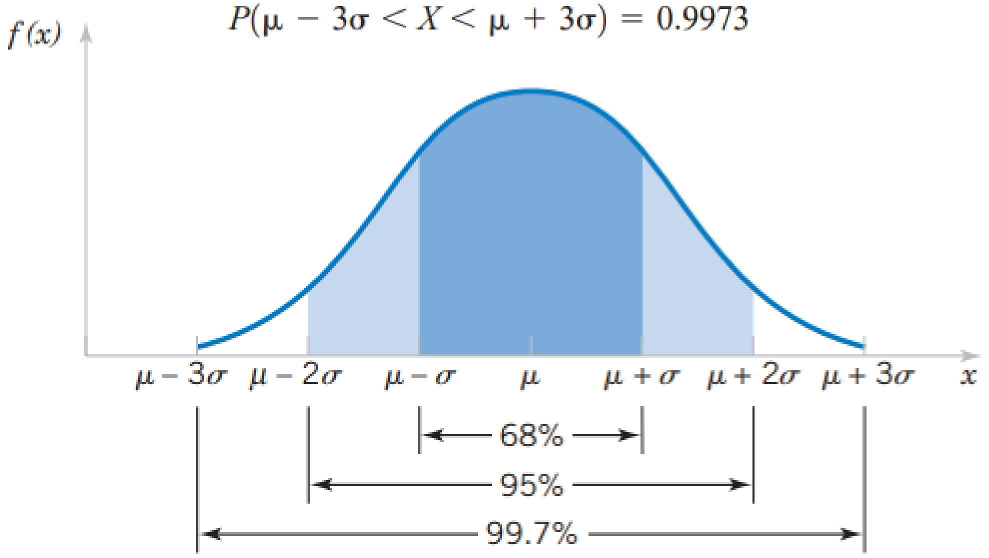
and the notation $N(\mu, \sigma^2)$ is used to denote the distribution.

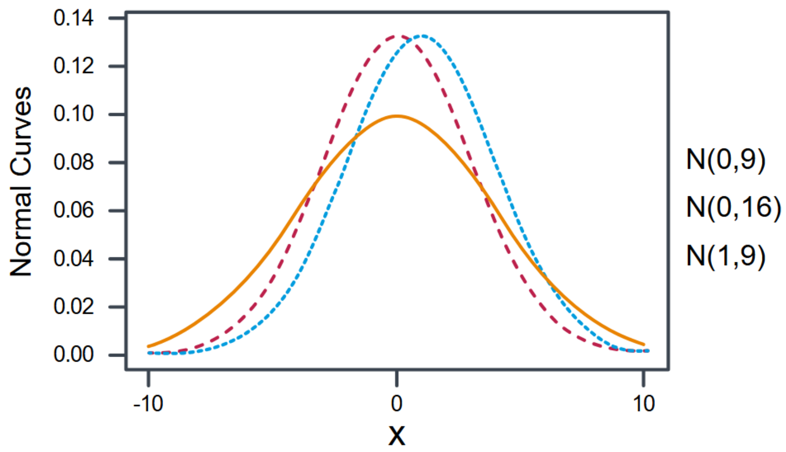
Important distribution: Normal

Properties:

- All normal curves are **bell-shaped** with points of inflection at μ±σ
- All normal curves are **symmetric about the mean** μ
- The area under an entire normal curve is 1.
- All normal curves are positive for all x
- The limit of f(x) as x goes to infinity is 0, and the limit of f(x) as x goes to negative infinity is 0
- The height of any normal curve is maximized at x= μ

$$P(\mu - \sigma < X < \mu + \sigma) = 0.6827$$
$$P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.9545$$
$$P(\mu - 3\sigma < X < \mu + 3\sigma) = 0.9973$$





Important distribution: Normal

Exercice: Let X equal the IQ of a randomly selected American which can be represented by a normal distribution $N(100,16^2)$.

What is the probability that a randomly selected American has an IQ below 90?

Important distribution: Normal

Solution:

$$P(X \leq 90) = F(90) = \int_{-\infty}^{90} \frac{1}{16\sqrt{2\pi}} \exp\left\{-\frac{1}{2}\left(\frac{x-100}{16}\right)^2\right\} dx$$

However, it is not possible to integrate the normal p.d.f. You can only approximate the integral using numerical analysis techniques!!!

So, you can create a table with the numerical approximation **BUT** you will need a table for all the different normal distributions, which is an infinity.

What can do? We can standardize our distribution to N(0,1) and have only 1 table for this distribution!!!

Important distribution: Normal

Notes: It is useful to define the standardized normal distribution.

If $X \sim N(\mu, \sigma^2)$, then:

$$Z = \frac{X - \mu}{\sigma}$$

follows the $N(0, 1)$ distribution, which is called the **standardized** (or **standard**) **normal distribution.**

$$F(z) = \Phi(z) = P(Z \leq z) = P\left(\frac{X - \mu}{\sigma} \leq z\right)$$

Important distribution: Normal

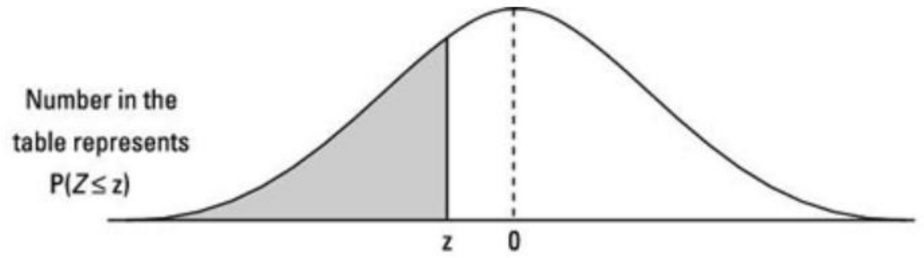
Exercice: Let X equal the IQ of a randomly selected American which can be represented by a normal distribution N(100,16).

What is the probability that a randomly selected American has an IQ below 90?

$z=(90-100)/16=-0.63$

So $P(X<90)=P(Z<-0.63)=0.2643$

What is the probability that a randomly selected American has an IQ **above** 140?



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
–3.6	.0002	.0002	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
–3.5	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002
–3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
–3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
–3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
–2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
–2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
–2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
–2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
–0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
–0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
–0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
–0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776

Important distribution: Normal

Solution: Let X equal the IQ of a randomly selected American which can be represented by a normal distribution N(100,16).

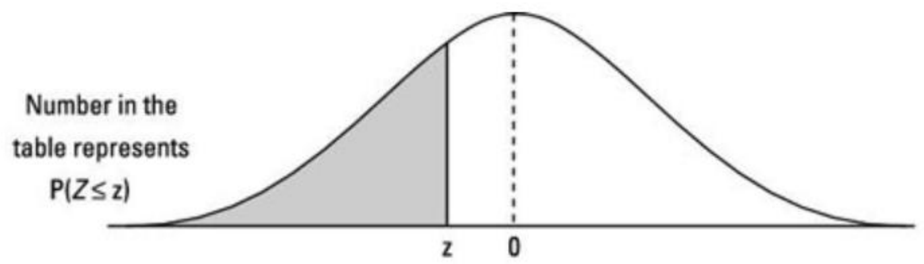
What is the probability that a randomly selected American has an IQ below 90?

z=(90-100)/16=-0.63

So P(X<90)=P(Z<-0.63)=0.2643

What is the probability that a randomly selected American has an IQ **above** 140?
z=(140-100)/16=2.5

P(X>140)=P(Z>2.50)=P(Z<-2.5)=0.0062



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
–3.6	.0002	.0002	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
–3.5	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002
–3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
–3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
–3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
–2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
–2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
–2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
–2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
–0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
–0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
–0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
–0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776

Important distribution: Normal

Exercice: Suppose *X*, the grade on a midterm exam, is normally distributed with mean 70 and standard deviation 10. The instructor wants to give 15% of the class an A.

What cutoff should the instructor use to determine who gets an A?

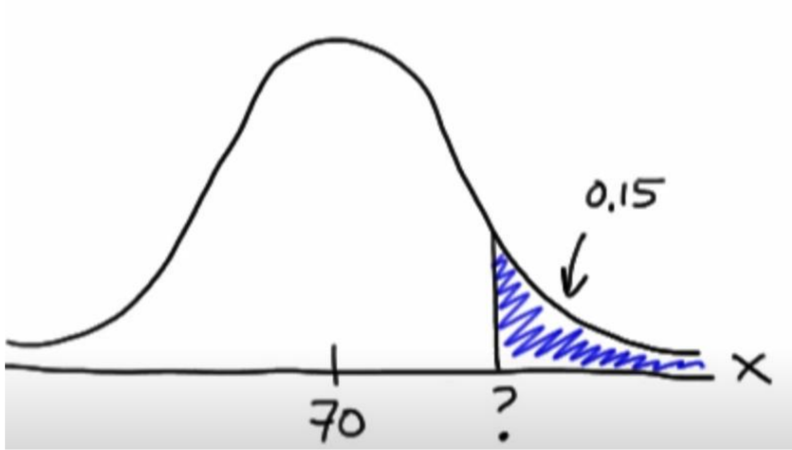
Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.0	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147

Important distribution: Normal

Solution: Suppose *X*, the grade on a midterm exam, is normally distributed with mean 70 and standard deviation 10. The instructor wants to give 15% of the class an A.

What cutoff should the instructor use to determine who gets an A?

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.0	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147



You need to find the z value where the probability of P(Z>z)=1-0.1500=0.8500

From this this table you see that z=1.04

z=(X-70)/10 therefore X=10.4+70=80.4

Important distribution: Normal

Exercice: Assume that a random variable is normally distributed with a mean of 24 and a standard deviation of 2. Consider an interval of length one unit that starts at the value a so that the interval is [a, a +1].

For what value of a is the probability of the interval greatest?

Important distribution: Normal

Solution: Assume that a random variable is normally distributed with a mean of 24 and a standard deviation of 2. Consider an interval of length one unit that starts at the value a so that the interval is [a, a +1].

For what value of a is the probability of the interval greatest?

Graphically is μ-1/2

Important distribution: Normal

Exercice: The length of stay at a specific emergency department in Paris, had a mean of 4.6 hours with a standard deviation of 2.9. Assume that the length of stay is normally distributed.

(a) What is the probability of a length of stay greater than 10 hours?

(b) What length of stay is exceeded by 25% of the visits?

(c) From the normally distributed model, what is the probability of a length of stay less than zero hours

Important distribution: Normal

Solution: The length of stay at a specific emergency department in Paris, had a mean of 4.6 hours with a standard deviation of 2.9. Assume that the length of stay is normally distributed.

(a) What is the probability of a length of stay greater than 10 hours?

$z=(10-4.6)/2.9=1.86$ therefore, $P(X>10)=P(Z>1.86)=1-0.9686=0.0314$

(b) What length of stay is exceeded by 25% of the visits?

$P(Z>z)=0.25$ therefore $z=0.67$, so $x=(0.67)*2.9+4.6=6.54$ hours

(c) From the normally distributed model, what is the probability of a length of stay less than zero hours

$P(X<0)=P(Z<-4.6/2.9)=P(Z<-1.586)=0.606$ so 6.1%

Important distribution: Exponential

Definition:

The random variable X that equals the distance between successive events of a Poisson process with mean number of events $\lambda > 0$ per unit interval is an **exponential random variable** with parameter λ. The probability density function of X is

$$f(x) = \lambda e^{-\lambda x} \quad \text{for} \quad 0 \leq x < \infty$$

If the random variable X has an exponential distribution with parameter λ,

$$\mu = E(X) = \frac{1}{\lambda} \quad \text{and} \quad \sigma^2 = V(X) = \frac{1}{\lambda^2}$$

Important distribution: Exponential

Example: In a large corporate computer network, user log-ons to the system can be modeled as a Poisson process with a mean of 25 log-ons per hour. What is the probability that there are no log-ons in an interval of 6 minutes?

Important distribution: Exponential

Solution: In a large corporate computer network, user log-ons to the system can be modeled as a Poisson process with a mean of 25 log-ons per hour. What is the probability that there are no log-ons in an interval of 6 minutes?

Let X denote the time in hours from the start of the interval until the first log-on. Then, X has an exponential distribution with λ=25 log-ons per hour

So 6min=0.1 hour

$$f(x) = \lambda e^{-\lambda x}$$

P(X>0.1)= $\int_{0.1}^{\infty} f(x)dx = \int_{0.1}^{\infty} 25 * e^{-25x}dx$ =-[$e^{-\infty}$- $e^{-2.5}$]=0.082

Important distribution: Exponential

Example: The number of miles that a particular car can run before its battery wears out is exponentially distributed with an average of 10,000 miles. The owner of the car needs to take a 5000-mile trip. What is the probability that he will be able to complete the trip without having to replace the car battery?

Important distribution: Exponential

Solution: The number of miles that a particular car can run before its battery wears out is exponentially distributed with an average of 10,000 miles. The owner of the car needs to take a 5000-mile trip. What is the probability that he will be able to complete the trip without having to replace the car battery?

At first glance, it might seem that a vital piece of information is missing. It seems that we should need to know how many miles the battery in question already has on it before we can answer the question

The exponential distribution is "**memoryless**." Therefore, the probability that the car battery wears out in more than y=5000 miles doesn't matter if the car battery was already running for x=0 miles or x=2000 miles

$$f(x) = \lambda e^{-\lambda x}$$

Therefore,

E[X]=10000=1/ λ

P(X>5000)= $\int_{5000}^{\infty} 1/10000 * e^{-x/10000} dx = e(-\frac{10000}{5000}) = 0.604$

Important distribution: Gamma

Definitions:

The **gamma function** is

$$\Gamma(r) = \int_0^\infty x^{r-1} e^{-x}\, dx, \quad \text{for } r > 0$$

The random variable X with probability density function

$$f(x) = \frac{\lambda^r x^{r-1} e^{-\lambda x}}{\Gamma(r)}, \quad \text{for } x > 0$$

has a **gamma random variable** with parameters $\lambda > 0$ and $r > 0$. If r is an integer, X has an Erlang distribution.

If X is a **gamma random variable** with parameters λ and r,

$$\mu = E(X) = r/\lambda \quad \text{and} \quad \sigma^2 = V(X) = r/\lambda^2$$